

Mostly rational types

Andrew Tai*

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Abstract

This paper proposes a method combining revealed preference theory and machine learning ideas to partition consumers in a principled way. The Generalized Axiom of Revealed Preference can be applied to partition consumers into fully rational types, in the classical consumer theory sense. From here, I construct a schedule between parsimony and rationality: for each number of types, households are partitioned into groups that are as close to rational as possible, in the sense of the CCEI or Varian indexes. I illustrate using household milk purchase data from the public-use FoodAPS survey. In the example data, 15 types are required to fully rationalize the data. However, many of these have few households, and far fewer types can describe the data at little cost to rationality. My method thus establishes a theory-driven and principled way to describe consumer demand data and the tradeoff between parsimony and heterogeneity.

1 Introduction

Gary Becker famously advocated for an approach to economics embracing assumptions of maximizing behavior, stable preferences, and equilibrium:

The assumption of stable preferences provides a stable foundation for generating predictions about responses to various changes, and prevents the analyst from succumbing to the temptation of simply postulating the required shift in preferences to 'explain' all apparent contradictions to his predictions. [Becker, 1976](#).

Simply, if economics is a science, then unexplained phenomena must not be consistently hand-waved away as unique phenomena. [Stigler and Becker \(1977\)](#) are even more stark:

Tastes neither change capriciously nor differ importantly between people.

*US Dept. of War. Contact: andrew.a.tai@gmail.com. Disclaimer: The views expressed are those of the author and do not reflect the official policy or position of the Department of War or the U.S. Government.

However, this poses a problem for the estimation of consumer behavior, where significant individual heterogeneity is natural. While some of this heterogeneity can be modeled on the consumers' observables (e.g. wealth, demographics, etc.), much is an accident of nature. Consider for example household milk purchases, where preference between milk categories is unlikely to be closely tied to observable characteristics.

Given consumption data from a panel of households, the classic approach is to pool the data, and control on household observables if available. An early example, [Banks, Blundell, and Lewbell \(1997\)](#), finds that the approach works well for some categories of purchases but very poorly for others. They apply the (quadratic) almost ideal demand system, which explicitly assumes a representative consumer. More recent models like [Berry, Levinsohn, and Pakes \(2004\)](#) allow random coefficients to incorporate individual heterogeneity (among other desiderata). Nevertheless, when households have significant unobservable heterogeneity, the resulting estimates may be imprecise; and point estimates may not accurately describe any households at all.

[Crawford and Pendakur \(2013\)](#) propose to apply revealed preferences, chiefly [Afriat \(1967\)](#), to partition households into fully rational types. Afriat's theorem establishes the Generalized Axiom of Revealed Preference (GARP) as a necessary and sufficient condition for a set of consumption data to be generated by a well-behaved utility function $u(x)$. Given an observed choice x_i at a price p_i , it must dominate any other affordable choice. The following is an example of a violation:

- Bundle x_1 was chosen at price p_1
- Bundle x_2 was chosen at price p_2
- $p_1 x_1 \geq p_1 x_2$ – that is, x_2 was affordable when x_1 was chosen
- $p_2 x_2 \geq p_2 x_1$ – that is, x_1 was affordable when x_2 was chosen
- At least one of the inequalities is strict

This violation implies a cycle in preferences, $x_1 \succeq x_2$ and $x_2 \succeq x_1$, with at least one strict. GARP generalizes this idea from 2-cycles to general length.

GARP is usually interpreted as a check on repeated observations from one individual (household). Crawford and Pendakur propose to use GARP to partition household data into rational types. That is, they ask *how many distinct utility functions are generating the consumer demand data?* That is, how many different utility functions exist in the population?

I apply this method to data from the USDA, National Household Food Acquisition and Purchase Survey (FoodAPS). I find that 15 types are necessary to rationalize the 2,167 households. However, many of these types contain very few households. I thus construct a schedule between the number of types and rationality: for each number of types k , households are partitioned into k types by a greedy algorithm that places each household so as to keep the partition as rational as possible, with rationality measured by either the Critical Cost Efficiency Index (CCEI) or the Varian index (cf. [Afriat \(1973\)](#) and [Varian \(1990\)](#)). In particular, use of the Varian index was

computationally impractical until recent work by [Demuynck and Rehbeck \(2023\)](#). The schedule makes the tradeoff explicit: sacrificing a modest amount of rationality (interpretable as unmodeled individual heterogeneity) supports far fewer types. My method thus offers intermediate options between two extremes – assuming all households have the same underlying utility; or requiring full rationality. By sacrificing a modest amount of rationality (interpretable as requiring unobserved individual heterogeneity), we require far fewer types.

The next section describes the dataset. Section 3 describes the proposed method to partition the households into groups. Section 4 estimates demand via almost ideal demand system (AIDS). Section 5 concludes.

2 Data

To illustrate my proposed method, I use data from the USDA FoodAPS survey, which was fielded from April 2012 through mid-January 2013. Each participating household reported the foods that household members acquired over a 7-day period. I focus particularly on household milk purchases, partitioned into four types of milk: whole, 2%, 1%, and skim/nonfat milk. Among the survey sample, 2,167 households are observed purchasing any milk; this forms my sample for analysis.

For each household, I observe the quantity of each type of milk purchased and the prices at which they were purchased. Prices were generally not recorded for goods not purchased; thus I impute prices when not recorded by the median in the matching category: region, rural / urban status, and store type. While FoodAPS level of observations is a purchase event (e.g. one receipt), I aggregate each household to a weekly sum. Table 1 summarizes the data. Note the extreme outlier shown in the maximum price of whole milk. I chose to leave this observation in the dataset, as the objective of this paper is to illustrate my proposed method, rather than to accurately estimate milk demand.

This paper does not seek to argue that estimating household milk demand is important in and of itself. Rather, this setting readily illustrates the issues that this paper seeks to note. While it is entirely possible to estimate a demand model on household milk purchases, the presence of individual heterogeneity would make difficult. The next section makes this clear. There may be individual heterogeneity by taste for milk (e.g. whole vs. skim) or by price sensitivity (e.g. some households are more price sensitive, or more likely to substitute between milk types). Even observing household characteristics, it may be difficult to describe this heterogeneity – much of it is likely due to unobservable taste.

3 Method

I now turn to a proposed method for partitioning households into distinct types. Based on the observed consumption data, the initial exercise prescribed by [Crawford and Pendakur \(2013\)](#) is

Table 1: Descriptive statistics ($N = 2167$)

	Mean	Min	Max	SD
<i>Budget shares</i>				
Whole	0.29	0.00	1.00	0.43
2%	0.46	0.00	1.00	0.47
1%	0.10	0.00	1.00	0.29
Skim	0.13	0.00	1.00	0.33
<i>Total expenditure (\$)</i>				
Total expenditure	4.97	0.30	88.47	4.30
<i>Prices (\$ per 100 g)</i>				
Whole	0.14	0.02	14.36	0.53
2%	0.10	0.02	0.79	0.06
1%	0.10	0.01	0.75	0.06
Skim	0.11	0.02	1.12	0.08

to partition households into groups, each of which satisfies GARP. That is, each group can be treated as having a single utility function. This enables us to interpret the errors from any demand estimation model purely as model misspecification error rather than individual heterogeneity.

Definition 1. Let $\{(x_i, p_i)\}_{i \in N}$ be observed consumption bundles and prices. Bundle x_i is *directly revealed preferred* to a bundle x , written $x_i R^0 x$, if $p_i \cdot x_i \geq p_i \cdot x$; it is *strictly directly revealed preferred*, $x_i P^0 x$, if $p_i \cdot x_i > p_i \cdot x$. The revealed preference relation R is the transitive closure of R^0 over the observed bundles.

The data satisfy the *generalized axiom of revealed preference* (GARP) if for all $i, j \in N$: $x_i R x_j$ implies not $x_j P^0 x_i$.

The core of this method is Afriat's theorem.

Theorem 1. For a set of consumer demand data $\{(x_i, p_i)\}_{i \in N}$, the following are equivalent: (i) there exists a continuous, concave, strictly increasing utility function u that rationalizes the data, i.e. $u(x_i) \geq u(x)$ for every i and every bundle x with $p_i \cdot x \leq p_i \cdot x_i$; (ii) the data satisfy GARP; (iii) there exist numbers U_i and $\lambda_i > 0$ such that $U_j \leq U_i + \lambda_i p_i \cdot (x_j - x_i)$ for all $i, j \in N$.

I refer the reader to [Afriat \(1967\)](#) for details (or any number of lecture notes available online; e.g. by [Echenique](#)). Crawford and Pendakur propose an infeasible algorithm, Algorithm 1, to partition the data into the fewest number of groups, each of which satisfies GARP.

Algorithm 1 Exact minimum-type partition

Require: Observations $I = \{1, \dots, n\}$

Ensure: Minimum number of types N^* and a rationalizing partition $\mathcal{G}^*$

```
1: for  $K = 1, \dots, n$  do
2:   for each partition  $\mathcal{G} = \{G_1, \dots, G_K\}$  of  $I$  into  $K$  nonempty sets do
3:     if  $G_k$  satisfies GARP for every  $k = 1, \dots, K$  then
4:        $N^* \leftarrow K$ 
5:        $\mathcal{G}^* \leftarrow \mathcal{G}$ 
6:       return  $(N^*, \mathcal{G}^*)$ 
7:     end if
8:   end for
9: end for
```

This algorithm is infeasible, since it requires checking GARP for every partition of the dataset. Thus it requires checking GARP a combinatorial number of times; even though GARP itself can be checked in $O(N^3)$ time.

Algorithm 2 calculates a feasible upper bound on the number of types. The observations are first randomly ordered. The first observation forms the first type. The next observation is checked for GARP consistency with the first type. If they are consistent, then it is added to this type; otherwise it is added to a new type. Each subsequent observation undergoes the same procedure. The entire process can be repeated multiple times, and the best (smallest) partition is chosen.

Algorithm 2 Crawford–Pendakur upper-bound algorithm

Require: Observations $I = \{1, \dots, n\}$ and number of restarts R

Ensure: Upper bound $\bar{N}$ and a rationalizing partition $\bar{\mathcal{G}}$

```
1:  $\bar{N} \leftarrow \infty$ 
2: for  $r = 1, \dots, R$  do
3:   Draw a random ordering  $(i_1, \dots, i_n)$  of  $I$ 
4:    $\mathcal{G} \leftarrow \emptyset$ 
5:   for  $t = 1, \dots, n$  do
6:      $\mathcal{C} \leftarrow \{G \in \mathcal{G} : G \cup \{i_t\} \text{ satisfies GARP}\}$ 
7:     if  $\mathcal{C} \neq \emptyset$  then
8:       Choose  $G^* \in \arg \max_{G \in \mathcal{C}} |G|$ 
9:       Replace  $G^*$  by  $G^* \cup \{i_t\}$  in  $\mathcal{G}$ 
10:    else
11:       $\mathcal{G} \leftarrow \mathcal{G} \cup \{\{i_t\}\}$ 
12:    end if
13:  end for
14:  if  $|\mathcal{G}| < \bar{N}$  then
15:     $\bar{N} \leftarrow |\mathcal{G}|$ 
16:     $\bar{\mathcal{G}} \leftarrow \mathcal{G}$ 
17:  end if
18: end for
19: return  $(\bar{N}, \bar{\mathcal{G}})$ 
```

Figure 1 shows Algorithm 2 applied to the FoodAPS data with 10 iterations. Over half of households can be described by a single utility function; there are 5-10 sizable groups, and the remainder are very small. Table 2 shows summary statistics for milk purchases by group; the descriptive statistics already illustrate significant heterogeneity in behavior across types. For example, Type 1 favors whole milk, Type 2 favors 2% milk, and Type 3 almost exclusively purchases 1% milk.

Motivated by this pattern, we may ask how much we gain in terms of “rationality” by having these small groups. Stated another way, do we lose much by having fewer groups? Doing comes with the practical advantage of allowing us to estimate fewer demand models; and the philosophical advantage of respecting Stigler’s and Becker’s charge of limiting capricious changes in taste.

Of course, by having fewer groups, some observations will not be rationalizable. That is, either some households are behaving irrationally, or have unexplained individual heterogeneity. A number of indexes exist to quantify the degree of irrationality. In this paper, I use the critical cost efficiency index (CCEI) and the related Varian index. Intuitively, CCEI asks *how much would we need to shrink every observed budget before the consumer’s choices become consistent with utility maximization?* A $CCEI = 1$ is completely rational; a $CCEI = 0.95$ is rational after allowing up

Figure 1: Partition of FoodAPS households into fully rational types

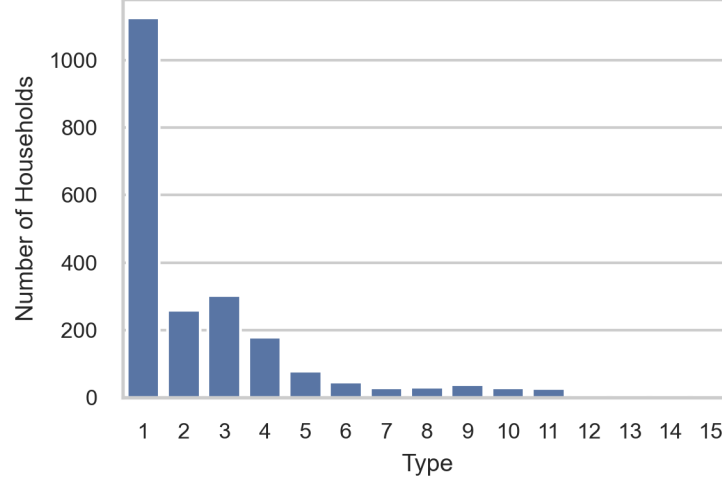


Table 2: Descriptive statistics by group

	<i>N</i>	Whole	2%	1%	Skim
Pooled	2167	0.29	0.47	0.10	0.14
Type 1	1125	0.23	0.57	0.09	0.12
Type 2	302	0.62	0.13	0.14	0.11
Type 3	259	0.03	0.95	0.02	0.00
Type 4	180	0.44	0.10	0.17	0.29
Type 5	78	0.40	0.07	0.11	0.42
Type 6	47	0.20	0.62	0.03	0.16
Type 7	39	0.88	0.07	0.02	0.03
Type 8	31	0.02	0.03	0.00	0.95
Type 9	29	0.59	0.03	0.34	0.04
Type 10	29	0.04	0.94	0.00	0.02
Type 11	28	0.01	0.11	0.68	0.19
Type 12	9	0.04	0.05	0.86	0.05
Type 13	7	0.84	0.04	0.00	0.12
Type 14	2	0.00	0.80	0.00	0.20
Type 15	2	0.00	0.00	0.73	0.27

to 5% of each household's budget to be wasted. The Varian index has us shrink each consumer's budget individually (rather than applying a universal multiplier), so is a gentler measure.

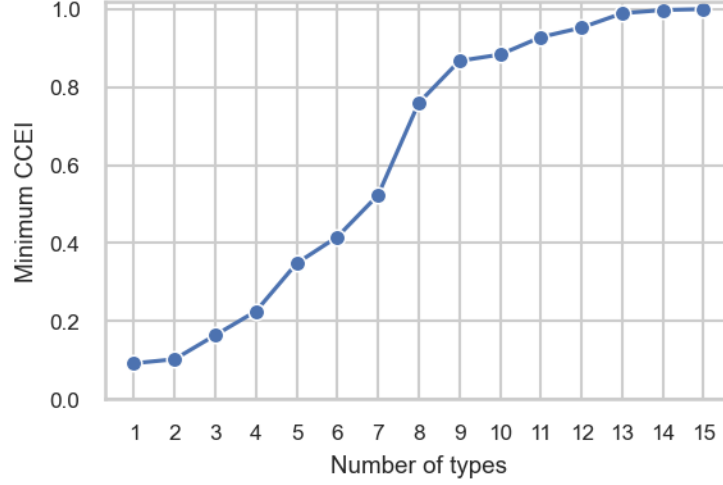
Definition 2. For $e \in [0, 1]$, define the relaxed relations $x_i R_e^0 x$ if $e p_i \cdot x_i \geq p_i \cdot x$ and $x_i P_e^0 x$ if $e p_i \cdot x_i > p_i \cdot x$, with R_e the transitive closure of R_e^0 . The *critical cost efficiency index* is $CCEI = \sup\{e \in [0, 1] : \text{the data satisfy GARP under } (R_e, P_e^0)\}$.

For a vector $e = (e_1, \dots, e_N) \in [0, 1]^N$, define the relaxed relations $x_i R_e^0 x$ if $e_i p_i \cdot x_i \geq p_i \cdot x$ and $x_i P_e^0 x$ if $e_i p_i \cdot x_i > p_i \cdot x$, so that each observation carries its own efficiency multiplier. The *Varian index* is $\sup\{\frac{1}{N} \sum_i e_i : \text{the data satisfy GARP under } (R_e, P_e^0)\}$. Setting $e_1 = \dots = e_N$ recovers the CCEI, so the Varian index is never below it.

I now turn to my proposed algorithm, which produces a schedule between the number of types and rationality. For a set of households G , let $e_G \in [0, 1]$ denote its CCEI when all of G is treated as one type. Define the rationality of a partition $\mathcal{P} = \{G_1, \dots, G_k\}$ as the minimum group CCEI, $\min_{G \in \mathcal{P}} e_G$: the common efficiency level at which every group simultaneously passes the relaxed GARP test.

For each number of types k , I find the highest efficiency level at which k types suffice. For a level e , relax GARP as in the CCEI's definition and run Crawford and Pendakur's upper bound algorithm under the relaxed relations (keeping the best of ten random orders): this bounds the number of types needed at that level. The number of types needed weakly rises with e , so the highest feasible level for each k is found by binary search over the finite set of candidate levels. Finally, each group's CCEI in the chosen partition is computed exactly, and the reported rationality is their minimum, which is at least the search level; the fully rationalizing 15-type partition guarantees full rationality at $k = 15$. The schedule is weakly increasing in the number of types by construction, since the set of feasible levels only grows as more types are allowed. I interpret lower rationality as requiring more unobserved individual heterogeneity to rationalize the data. Algorithm 3 summarizes the procedure; the Varian variant instead assigns each household the largest multiplier consistent with its group, as described below.

Figure 2: Schedule of CCEI – Types



Algorithm 3 Rationality schedule by efficiency search (CCEI measure)

Require: Observations $I = \{1, \dots, n\}$; maximum types K ; restarts R ;
candidate levels $\mathcal{E}$: the cost ratios in $(0, 1)$, and 1

Ensure: Rationality schedule $\rho(k)$ and a partition for each $k = 1, \dots, K$

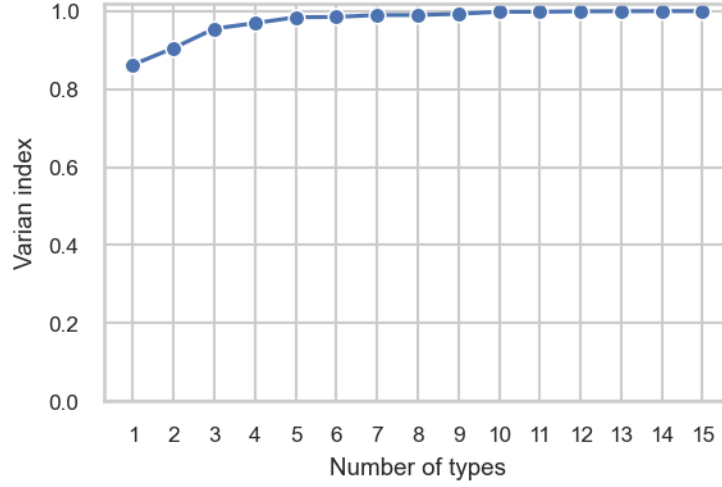
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1: function TYPES( $e$ )
2:   run Algorithm 2 with GARP relaxed to level  $e$ , for  $R$  random orders
3:   return the fewest number of groups  $N(e)$  and its partition  $\mathcal{G}(e)$ 
4: end function
5: for  $k = 1, \dots, K$  do
6:    $e_k \leftarrow$  largest  $e \in \mathcal{E}$  with  $N(e) \leq k$ , by binary search over  $\mathcal{E}$ 
7:   compute each group's CCEI in  $\mathcal{G}(e_k)$ ;  $\rho(k) \leftarrow$  their minimum
8:   if  $k > 1$  and  $\rho(k) < \rho(k - 1)$  then
9:      $\rho(k) \leftarrow \rho(k - 1)$  and reuse the  $k - 1$  partition
10:  end if
11: end for

```

Figure 2 shows the schedule of rationality and types. At 15 types, the data are fully rationalized: every group has a CCEI of 1. Combining all households into one type results in significant irrationality (that is, significant individual heterogeneity). In between, the schedule shows exactly how much rationality each additional type buys, and we can reduce the number of types substantially while keeping every type close to rational. Informally, it is *approximately correct* to treat the households within each group as sharing the same utility function.

Figure 3: Schedule of Varian index – Types



I also repeat the analysis for the Varian index, which allows each household to waste an individual amount of its budget. Here each household carries its own efficiency multiplier e_i : when a household is assigned, it receives the largest multiplier consistent with its group, and existing members keep theirs. The reported rationality at each k is the simple average of the e_i across all households. The resulting measure is smoother and less punishing, since one badly-behaved household no longer drags down its whole group. Computing the Varian index exactly is NP-hard (see [Demuyne and Rehbeck \(2023\)](#) for an exact mixed-integer programming approach); the tracked multipliers are feasible by construction, so the reported index is a slightly conservative lower bound.

The Varian index is my favored measure of rationality. The CCEI schedule reports a worst-case statistic: because the multiplier is common to the whole group, a single hard-to-rationalize household caps the measured rationality of its entire group, and hence of the partition. The Varian index instead charges each household only for its own violations, so the schedule reflects the typical household rather than the most extreme one; and the average multiplier has a direct economic interpretation as one minus the average share of the milk budget a household would need to waste for its group’s utility function to explain its choices. I retain the CCEI schedule as a stringent worst-case bound. Figure 3 shows the schedule of Varian index rationality to number of types.

4 Demand estimation

Incorporating individual heterogeneity into demand estimation is difficult; see for example [Matzkin \(2003\)](#). However, the almost ideal demand system (AIDS) is easy to estimate, at the cost of assuming away individual heterogeneity. The previous section’s method offers a principled way to

quantify the size of this assumption.

To illustrate, I estimate AIDS models by type. For household i and milk type j , the share equation is

$$w_i^j = \alpha_j + \sum_{k=1}^4 \gamma_{jk} \ln p_i^k + \beta_j \ln \left(\frac{x_i}{P_i} \right) + \varepsilon_i^j$$

where

$$\ln P_i = \alpha_0 + \sum_k \alpha_k \ln p_i^k + \frac{1}{2} \sum_k \sum_\ell \gamma_{kl} \ln p_i^k \ln p_i^\ell$$

and w_i^j is i 's share of the milk budget spent on j , and p_i^j is the price of milk j faced by household i . Prices and expenditures are normalized so that $\ln p^j = \ln x = 0$ at the medians. Restrictions are

- $\sum_k \alpha_k = 1$, $\sum_k \beta_k = 0$, and $\sum_j \gamma_{jk} = 0$ for all k – expenditure shares over milk categories sum to 1
- $\sum_k \gamma_{jk} = 0$ for all j – homogeneity of degree zero in prices and total expenditure
- $\gamma_{jk} = \gamma_{kj}$ – symmetry of the Slutsky matrix

The α_j are interpreted as predicted budget shares at the median prices and median expenditure. β_j is expenditure semi-elasticity of the share, $\partial w_j / \partial \ln x$; that is, the change in j 's share of the budget per log change in total milk expenditure. Intuitively, $\beta_j > 0$ implies milk j is a luxury, and $\beta_j < 0$ implies it is a necessity — in the conditional sense, relative to the household's milk budget rather than its total expenditure. γ_{kl} is the change in the share of k per unit change in the log of the price of l , $\ln p^l$, holding real expenditure fixed.

Each type's system is estimated by iterated restricted seemingly unrelated regressions, following [Deaton and Muellbauer \(1980\)](#). The tables report the levels α_j and expenditure semi-elasticities β_j with standard errors, omitting the price coefficients γ_{jk} for space.

Since the Varian index is the favored measure, I estimate demand on the Varian schedule's partition at 3 types — the smallest number of types whose average Varian index reaches 95%, Varian's own rule of thumb for essentially rational behavior. Table 3 shows the estimates. The pooled sample estimates are incorrect on theoretical grounds (since we know we cannot treat all households as sharing a common utility function), and in practical terms they mask considerable heterogeneity. In levels, Type 1 (58% of households) is the mainstream whole-and-2% type; Type 2 (30%) tilts toward the lower-fat varieties, with 1% and skim taking nearly twice their pooled shares; and Type 3 (12%) is polarized between whole and skim milk, buying almost no 2%. The expenditure responses are starker: the pooled semi-elasticities are all close to zero, but this is an artifact of averaging sign-opposed behavior. As milk spending rises, Type 1 shifts toward whole milk and away from 2% (semi-elasticities 0.19 and -0.24), while Type 2 does exactly the reverse (-0.16 and 0.50); Type 3 shifts modestly toward 1%. Converting to expenditure elasticities ($1 + \beta_j / w_j$, conditional on the milk budget), Type 1's elasticities are about 1.6 for whole milk and 0.6 for

2% — a luxury and a necessity — while Type 2’s whole-milk elasticity is approximately zero: its whole-milk quantity barely moves as its milk spending grows. A pooled model would conclude that expenditure barely matters for milk composition, when in fact it matters substantially — in opposite directions — for almost nine-tenths of the sample.

Table 3: AIDS levels and expenditure semi-elasticities by Varian-schedule type (all households, 3 types). Standard errors in parentheses.

	N	Whole	2%	1%	Skim
<i>Levels, a^j</i>					
Pooled	2167	0.288 (0.009)	0.471 (0.010)	0.100 (0.006)	0.141 (0.007)
Type 1	1252	0.306 (0.012)	0.560 (0.013)	0.066 (0.007)	0.069 (0.007)
Type 2	655	0.165 (0.012)	0.410 (0.013)	0.188 (0.014)	0.237 (0.015)
Type 3	260	0.523 (0.030)	0.052 (0.011)	0.084 (0.019)	0.340 (0.028)
<i>Expenditure semi-elasticities, b^j</i>					
Pooled	2167	-0.002 (0.015)	0.014 (0.016)	0.017 (0.010)	-0.029 (0.011)
Type 1	1252	0.194 (0.020)	-0.236 (0.022)	0.024 (0.011)	0.018 (0.012)
Type 2	655	-0.164 (0.021)	0.496 (0.024)	-0.106 (0.025)	-0.226 (0.026)
Type 3	260	-0.102 (0.032)	0.016 (0.012)	0.115 (0.019)	-0.029 (0.030)

5 Conclusion

I propose a method combining revealed preference theory and machine learning ideas to partition consumers in a principled way. The Generalized Axiom of Revealed Preference can be applied to partition consumers into fully rational types, in the classical consumer theory sense. From here, a schedule between parsimony and rationality can be traced: for any number of types, households are partitioned into groups that are as close to rational as possible.

In the example data, 15 types are required to fully rationalize household milk purchases; however, many of these have few households, and far fewer types describe the data without significantly harming rationality. My method thus establishes a theory-driven and principled way to describe consumer demand data and the tradeoff between parsimony and heterogeneity. This matters in practice: application of AIDS models shows significant heterogeneity between household types.

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